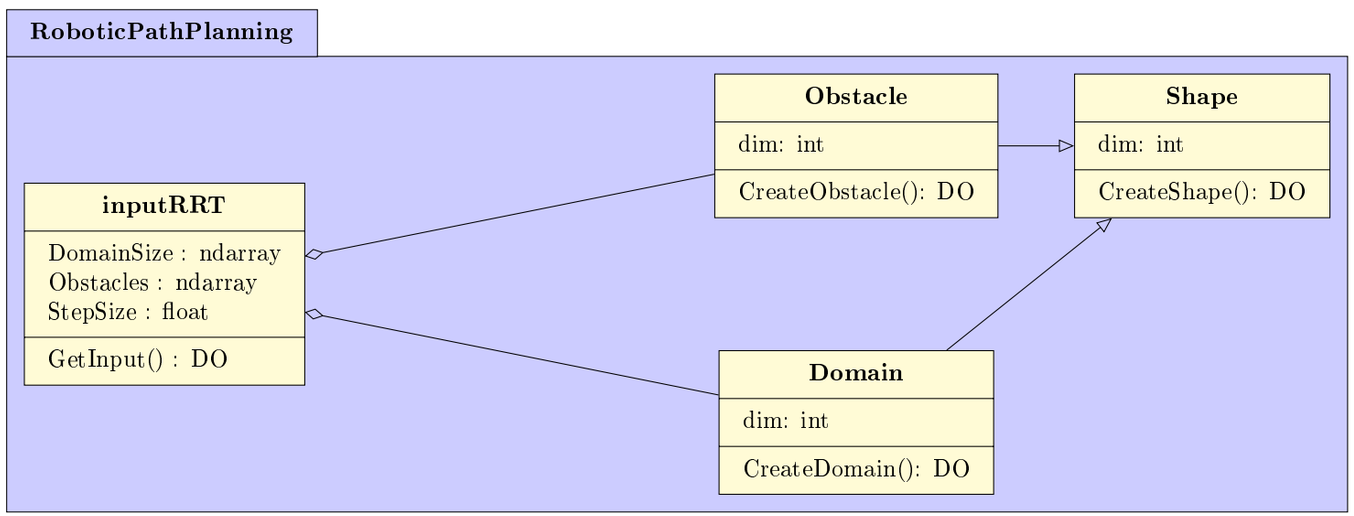
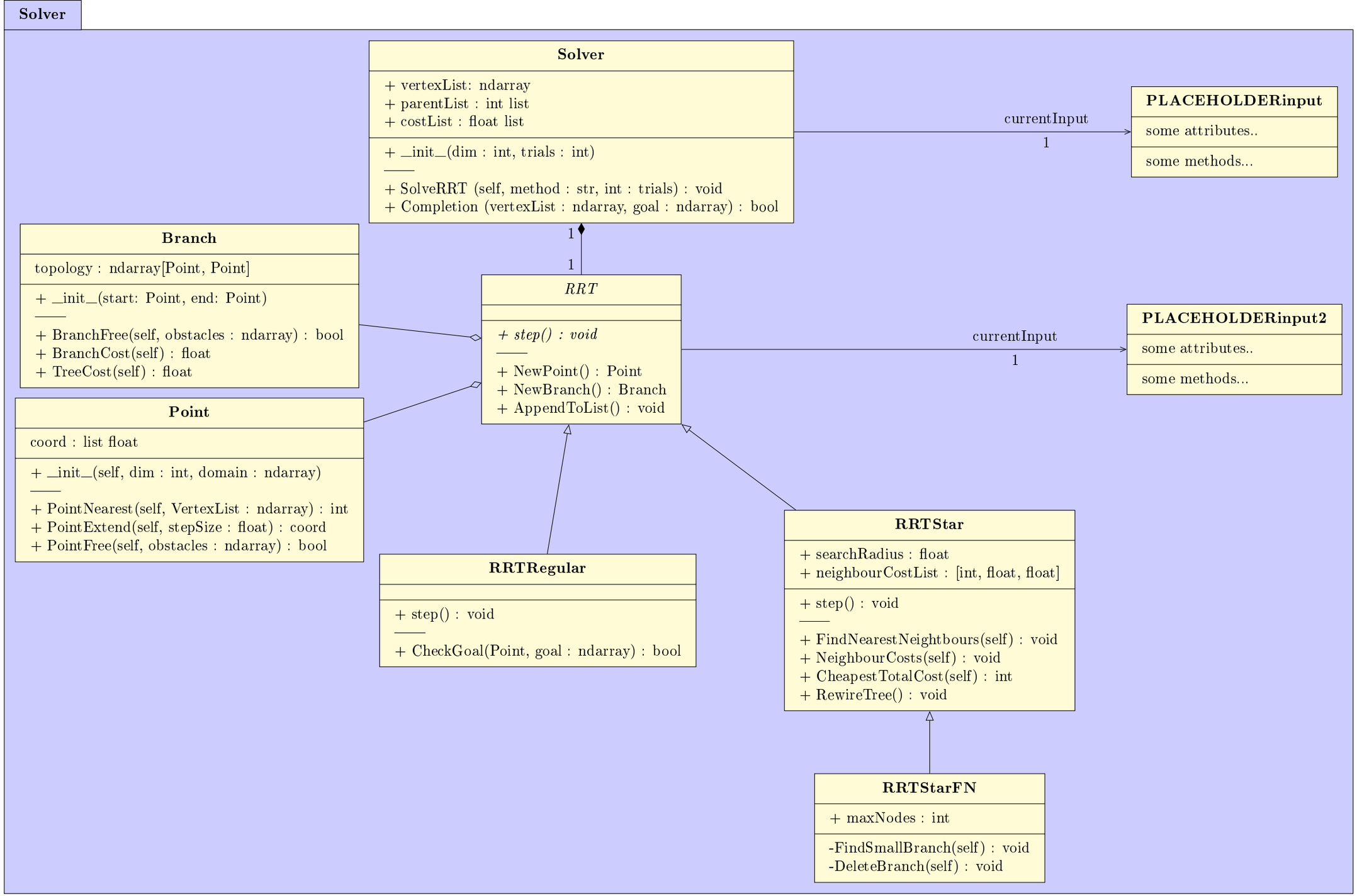
# UML Class Diagram for Robotic Path Planning

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Solver
Solver
+ vertexList: ndarray
+ parentList : int list
+ costList : float list
+ __init__(dim : int, trials : int)
+ SolveRRT (self, method : str, int : trials) : void
+ Completion (vertexList : ndarray, goal : ndarray) : bool
currentInput
1
PLACEHOLDERinput
some attributes..
some methods...
1
1
Branch
topology : ndarray[Point, Point]
+ __init__(start: Point, end: Point)
+ BranchFree(self, obstacles : ndarray) : bool
+ BranchCost(self) : float
+ TreeCost(self) : float
RRT
+ step() : void
+ NewPoint() : Point
+ NewBranch() : Branch
+ AppendToList() : void
currentInput
1
PLACEHOLDERinput2
some attributes..
some methods...
Point
coord : list float
+ __init__(self, dim : int, domain : ndarray)
+ PointNearest(self, VertexList : ndarray) : int
+ PointExtend(self, stepSize : float) : coord
+ PointFree(self, obstacles : ndarray) : bool
RRTRegular
+ step() : void
+ CheckGoal(Point, goal : ndarray) : bool
RRTStar
+ searchRadius : float
+ neighbourCostList : [int, float, float]
+ step() : void
+ FindNearestNeightbours(self) : void
+ NeighbourCosts(self) : void
+ CheapestTotalCost(self) : int
+ RewireTree() : void
RRTStarFN
+ maxNodes : int
-FindSmallBranch(self) : void
-DeleteBranch(self) : void